

CHAPTER – 10**SOLUTION AND COLLOIDS****"SOLUTION"****DEFINITION:**

A solution is a homogeneous mixture or single phase mixture of the molecules, atoms or ions of the two or more components "

EXPLANATION:

Particles of solute and solvent in a solution are extremely small having a diameter approximately of 1 nm. This indicates that;
Solution is a transparent mixture of its components.

Components of solution cannot be separated by filtration.
Light cannot be scattered through the solution. This is known as Tyndall effect.
There are basically nine types of solutions. Three gaseous, three liquids & three solid solutions.

"SUSPENSION"**DEFINITION:**

Suspension is a cloudy, heterogeneous mixture whose dispersed particles are settled down."

EXAMPLE:

If you fill glass with muddy water you will observe that mud particles will be settled down. This is due to the fact that solute particles are usually more than 1000 nm in size.

"HYDROPHILIC MOLECULES"**DEFINITION:**

The term **HYDRO** means water and **PHILIC** means to attract

Solute substances whose particles (ions/molecules) are strongly associated with water are classified into hydrophilic molecules."

"HYDROPHOBIC MOLECULES"**DEFINITION:**

HYDRO means water and **PHOBIC** means to dislike

"The molecules which are non-polar in nature due to this they do not have ability to attract water molecule."

NATURE OF SOLUTION IN LIQUID PHASE:

On the basis of solid, liquid and gaseous state there are nine types of solution.

i. Completely miscible liquids:

When two liquids are mixed into each other in all proportion to form a homogeneous mixture is called completely miscible liquid.

Example:

While studying on molecular level, the intermolecular attractions of the two liquids found to be of similar. Methanol and water both are polar molecules and on mixing they are completely soluble into each other.

ii. Partially miscible liquids:

Some Liquids dissolve into each other to a limited extent up to a certain concentration. When such liquids mix into each other, they form two separate phase where each phase is a saturated solution of liquid in which one serves as solute and other is solvent.

Example:

When phenol is mixed into water at 25°C , a two phase solution is formed in which upper layer represents 5 % phenol in 95 % water while lower layer consists of 70 % phenol in 30 % water.

iii. Completely immiscible liquids:

When two liquids do not mix into each other and do not form solution are called immiscible liquids.

Example:

Water is polar liquid and benzene is non polar. These two liquids are totally immiscible in all proportion and form two separate layers on mixing.

"SOLUBILITY"**DEFINITION:**

"Solubility is the amount of solute (in grams) that can be dissolved in a fixed quantity (volume) of a particular solvent to prepare a saturated solution at a specific temperature".

Solubility of a substance is fundamentally depends on chemical nature of solute and solvent. However, it notably changes with temperature and pressure.

FACTOR AFFECTING SOLUBILITY:**Effect of Temperature:****INCREASING TREND:**

Solubility of many solids in liquid increases with the rise of temperature. This is attributed with the fact that rise of temperature increases the kinetic energy and randomness of particles and create more spaces among solvent molecules in which solid solute particles can accommodate. Many solids dissolve into solvent with the absorption of heat, solubility of these substances increases with the rise in temperature.

DEACREASING TREND:

The dissolution of some solids (AlCl_3 , Na_2SO_4 and e.t.c.) and many liquids as well as gases in liquid solvent is carried out by exothermic process. Solubility of these substances decreases with the rise of temperature.

This dual behavior in the solubility of solids with the temperature can be predicted by whether heat absorb or evolve in the formation of solution

Effect of Pressure:

Solubility of solids and liquids in a liquid solvent is not appreciably change by altering the pressure. However, the solubility of gases into liquid is significantly affected by pressure. In theoretical aspect the dissolution of gases into liquid is an equilibrium process. This means that a simultaneous entrance and exist of gas molecules from solution occur. Thus an increases in pressure increases the number of

molecule so the rate of entrance of gas molecules into solution becomes higher than its exit which it in increase of solubility of gas into liquid with the rise of pressure.

"CONCENTRATION OF SOLUTION"

DEFINITION:

"The amount of solute in specific amount of solvent is known as the concentration".

UNITS OF CONCENTRATION:

Concentration of solution may be expressed in several physical and chemical units. Such as molarity, molality, normality, mole fraction etc.

"MOLAR CONCENTRATION (MOLARITY)"

DEFINITION:

"Numbers of moles of solute per cubic decimeter (dm^3) or per liter of solution is termed as Molarity"

It is denoted by "M". A solution is said to be molar solution if 1 mole of solute is dissolved in a dm^3 of a solution.

"MOLAL CONCENTRATION (MOLALITY)"

DEFINITION:

"Numbers of moles of solute per kilogram (1000 g) of solvent is termed as Molality".

It is also denoted by "m". A solution is said to be one molal if one mole of solute is dissolved in 1 kg of solvent.

"PERCENTAGE COMPOSITION"

"The mass percent (w/w %) may be define as the mass of solute in gram per 100 gram of solution."

12% (w/w) aqueous sugar solution means 12g of sugar dissolves in 100g solution or 88g of water.

"MOLE FRACTION (X)"

This unit is used when two or more components are present in the solution.

"It is the ratio of number of moles of one component to that of all components of solution."

The sum of mole fractions of all the components of a solution is always a unity.

$$X_A + X_B = 1$$

Example:

If a solution is made up of 1 mole methanol in 4 moles water, the mole fraction of methanol will be 0.2 and water will be 0.8. Now if we add the mole fraction of both methanol and water we get unity.

$$X_{\text{METHANOL}} + X_{\text{WATER}} = 0.2 + 0.8 = 1$$

Parts Per Million, Billion and Trillion:

These units are used in analytical or biochemical research works when solute is present in very small quantity.

"PARTS PER MILLION (PPM)"

"It is the number of parts or amount (mass or volume) of solute per million parts (10^6) of the amount (mass or volume) of solution."

$$\text{ppm} = \frac{\text{Amount of solute (mass or volume)}}{\text{Amount of solution (mass or volume)}} \times 10^6$$

"PARTS PER BILLION (PPB)"

"It is the number of parts or amount (mass or volume) of solute per billion parts (10^9) of the amount (mass or volume) of solution."

$$\text{ppb} = \frac{\text{Amount of solute (mass or volume)}}{\text{Amount of solution (mass or volume)}} \times 10^9$$

"PARTS PER TRILLION (PPT)"

"It is the number of parts or amount (mass or volume) of solute per trillion parts (10^{12}) of the amount (mass or volume) of solution."

"RAOULT'S LAW"**INTRODUCTION:**

A French chemist F.M. Raoult in 1887 studied the phenomena of lowering of vapours pressure by the effect of solute particles in a solution and formulated a law (1887) known as Raoult's law.

STATEMENT:

For a non-volatile solute system it states that:

"The Vapour pressure of a solution which is made up of a non-volatile solute with volatile solvent is directly proportional to the mole fraction of solvent".

MATHEMATICALLY:

$$\begin{aligned} P &\propto X_1 \\ P &= P^0 X_1 \text{ --- (i)} \end{aligned}$$

Here P specifies the vapours pressure of solution, P^0 represents vapours pressure of solvent and X_1 is the mole fraction of solvent.

Since in a binary solution;

$$\begin{aligned} X_1 + X_2 &= 1 \\ \text{Hence, } X_1 &= 1 - X_2 \end{aligned}$$

Putting the value of X_1 , in eq. (i) we get

$$\begin{aligned} P &= P^0 (1 - X_2) \\ P &= P^0 - P^0 X_2 \\ P^0 - P &= P^0 X_2 \\ \Delta P &= P^0 X_2 \text{ --- (ii)} \end{aligned}$$

Equation (ii) represents the second form of Raoult's law which states as the lowering of vapour pressure is directly proportional to the mole fraction of solute.

$$\Delta P / P^0 = X_2 \text{ --- (iii)}$$

Here $\Delta P / P^0$ is referred as relative lowering in vapours pressure.

Eq. (iii) represents third form of Raoult law which states as the relative lowering of vapours pressure is equal to the mole fraction of solute.

BINARY SOLUTION:

Raoult's law is also applicable on a binary solution made up of two miscible volatile liquids having similar order of polarity and molecular size. The partial pressure of each liquid in the mixture is equal to the vapours pressure of pure component multiplied by its mole fraction and may be written as:

$$P_A = P_A^\circ X_A \text{ and } P_B = P_B^\circ X_B$$

Thus the total pressure of solution is equal to the sum of partial pressure of both liquids in the mixture.

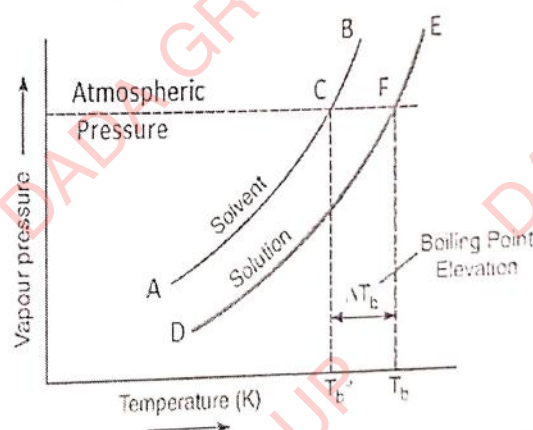
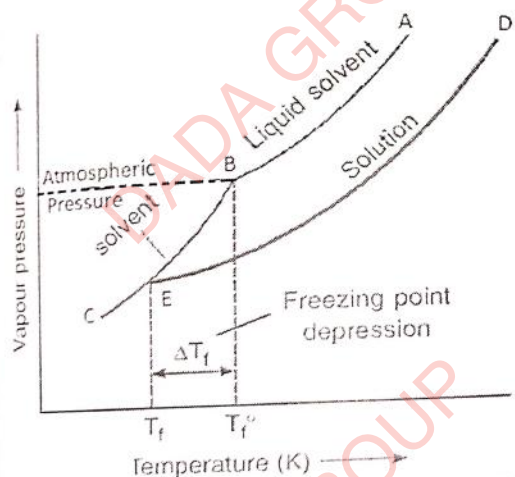
$$P_T = P_A^\circ X_A + P_B^\circ X_B \text{ --- (iv)}$$

"ELEVATION OF BOILING POINT"

"Boiling point of a solution of non-volatile solute is higher than pure solvent, this is a colligative property and known as elevation of boiling point (ΔT_b)".

The temperature at which vapours pressure of a liquid becomes equal to atmospheric pressure (1 atm) is known as normal boiling point. If a non-volatile and non-electrolyte solute (solid) is dissolved into a volatile solvent, its vapours pressure is found to be lower than 1 atm. Since liquid boils only when its vapours pressure becomes equal to atmospheric pressure, so more heat is needed to the solution to reach its vapours pressure to 1 atm.

It is concluded that the boiling point of solution is greater than the boiling point of pure solvent.



"DEPRESSION OF FREEZING POINT"

We know that kinetic energy of liquid molecules decreases on cooling and molecules approach one another more closely and liquid finally turns into solid.

"Freezing point of a liquid is the temperature at which liquid and solid phases co-exist in equilibrium with the same vapours pressure".

EXPLANATION:

The presence of a non-volatile solute lowers the freezing point of solvent. This can be explained in terms of vapours pressure. Since at freezing point the vapours pressure of both liquid and solid phases must be the same, the addition of solute causes a fall in vapours pressure. Thus in order to re-establish the equilibrium of two phases, temperature of solution is consequently lowered below the freezing point of pure solvent.

"OSMOSIS"

Surface of certain materials permit water and other smaller size particles to pass through but do not allow large solute particles. These serve as semi permeable membrane for example cellophane, animal bladder etc. Although the passage of solvent particles takes place in both directions of semi permeable membrane but their movement is faster from dilute solution to concentrated solution. This spontaneous process is commonly known as osmosis.

"OSMOTIC PRESSURE"

"The hydrostatic pressure exerted by the solution which stops the process of osmosis is called osmotic pressure".

"IDEAL SOLUTION"

"The solution which perfectly obey the Raoult's law are referred as ideal solutions"

Characteristic of an Ideal Solution:

- It obeys Raoult law at all temperature and concentration.
- The molecular interaction of liquid A-A, B-B and A-B remains almost same.
- No heat absorbs or evolve in the formation of ideal solution ($\Delta H = 0$).
- No change in the volume takes place in the formation of ideal solution ($\Delta V = 0$).

Methanol, Ethanol, Benzene-Toluene are form ideal solution on mixing.

"NON-IDEAL SOLUTION"

"Solutions which do not satisfy Raoult's law due to the difference in the attractive forces and molecular size of solute and solvent particles. These solutions are formed by the change in volume and enthalpy and known as non-ideal solution".

"COLLOIDS"

"Colloids is a mixture in which one substance consisting of microscopically dispersed insoluble particles in suspended."

EXAMPLE:

Milk is a homogenous mixture of tiny particles of casein and some other ingredients in aqueous phase. These particles neither settle down like in suspension nor serves as solute of a solution.

SIZE:

The size of colloidal particles is smaller than floating particles of suspension but larger than solute particles of solution. The size range of colloidal particle is from 1 to 1000nm.

TYNDALL EFFECT:

Unlike solution they have ability to scatter light and this property is known as Tyndall effect.

WATER FILTRATION PLANT:

Reverse osmosis works on the basis of passage of water molecules across a semi permeable membrane from impure water to pure water by applying a high pressure which overcomes the osmotic pressure. In other word flow of water molecules takes places in reverse direction as carried out by natural osmosis. About 95 to 99 percent of the dissolved contaminant like salts, metal ions, protozoa and e.t.c. are removed if water is processed by reverse osmosis.

SHORT QUESTIONS

1. Name various units of concentration and explain mole fraction.
2. Define Molarity and molality. Which of these depends on temperature?
3. Explain on particles bases how the vapours pressure of solution is lowered by adding non-volatile solute.
4. What are miscible and immiscible liquids? Why n-hexane (petrol) is immiscible in water?
5. Solubility of oxygen in water increases with pressure but solubility of glucose in water has negligible effect of pressure. why?

DESCRIPTIVE QUESTIONS

1. State Raoult's law and derive its mathematical expression.
2. What does an ideal solution means? Give four characteristics to distinguish between ideal and non-ideal solution.
3. Define osmosis and osmotic pressure. Give four daily life examples of osmosis.
4. What are colligative properties? Why does the boiling point of a liquid get raised when a non-volatile solute is added?
5. Differentiate among true solution, colloidal solution and suspension on the bases of:
 - (i) Particle size
 - (ii) Visibility

NUMERICAL QUESTION

1. Automotive antifreeze is a 60% (w/w) aqueous solution of ethylene glycol ($C_2H_6O_2$). Determine (a) molality of solution (b) mole fraction of ethylene glycol in the solution.
[Answer: molality = 24.19mol/kg, $X = 0.3$]
2. A solution is prepared by mixing 46g ethanol (C_2H_5OH) and 180g water. Calculate the mole fraction of both components.
[Answer: $X_1 = 0.090$; $X_2 = 0.909$]

CHAPTER# 11"THERMOCHEMISTRY""THERMOCHEMISTRY"

DEFINITION:

"It is the branch of physical chemistry which deals with the measurement of heat evolved or absorbed during a chemical reaction."

"THERMODYNAMICS"

DEFINITION:

"It is a branch of science which deals with the study based on the principle of conservation of energy."

UNIT OF HEAT:

In S.I system the unit of heat is Joule. The relation b/w joule & calorie is given below:

$$1 \text{ Joule} = 0.239 \text{ Cal} \quad \text{"OR"} \quad 1 \text{ Cal} = 4.184 \text{ Joule}$$

"THERMOCHEMICAL REACTIONS"

DEFINITION:

Chemical reactions, which are accompanied by energy changes as well as with the material changes, are generally known as "Thermo chemical reactions".

TYPES OF THERMOCHEMICAL REACTIONS:

There are two types of thermo chemical reactions.

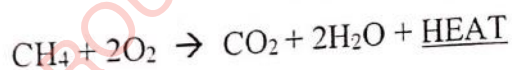
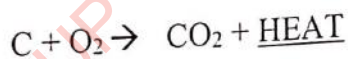
- 1- Exothermic reactions.
- 2- Endothermic reactions.

"EXOTHERMIC REACTION"

DEFINITION:

The chemical reactions which are accompanied by the liberation or emission of energy are called "Exothermic reactions".

EXAMPLE:



EXPLANATION:

All combustion reactions are exothermic reactions. In exothermic reaction, the enthalpy of the reactant (H_1) is greater than the enthalpy of product (H_2).

Therefore, in exothermic reactions ΔH is negative.

$$\Delta H = H_2 - H_1$$

$$\Delta H = -ve$$

“ENDOTHERMIC REACTIONS”**DEFINITIONS:**

The chemical reactions are accompanied by the absorption of energy are called “Endothermic reactions”.

EXAMPLE:**EXPLANATION:**

In endothermic reactions, the enthalpy of product (H_2) is greater than enthalpy of reactant (H_1).

Therefore, in endothermic reactions ΔH is positive.

$$\Delta H = H_2 - H_1$$

$$\Delta H = +ve$$

“DIFFERENCE”

<u>ENDOTHERMIC REACTIONS</u>	<u>EXOTHERMIC REACTIONS</u>
➤ The chemical reactions are accompanied by the absorption of energy are called “ <u>Endothermic reactions</u> ”.	➤ The chemical reactions which are accompanied by the liberation or emission of energy are called “ <u>Exothermic reactions</u> ”.
➤ In these reactions, the enthalpy of product (H_2) is greater than enthalpy of reactant (H_1).	➤ In these reactions, the enthalpy of the reactant (H_1) is greater than the enthalpy of product (H_2).
➤ In these reactions ΔH is +ve.	➤ In these reactions ΔH is -ve.
➤ In these reactions heat is reactant.	➤ In these reactions heat is product.
➤ In these reactions temperature of the mixture is decreased.	➤ In these reactions temperature of the mixture is increased.

"FACTORS AFFECTING AMOUNT OF HEAT"

Heat evolved or absorbed during chemical reactions depends upon:

- The amount of chemical substances involved.
- The physical state of the substances involved.
- Temperature.
- Whether the reaction occurs at constant pressure or constant volume.

"THERMODYNAMIC TERMS"

"SYSTEM"

DEFINITION:

"Any real or imaginary portion of the universe or anything which is under observation/consideration is called system."

TYPES OF SYSTEM:

On the basis of their conductive behavior:

There are three types of system.

- Open system.
- Closed system.
- Isolated system.

● OPEN SYSTEM:

"A system in which matter as well as energy can be exchanged to the surrounding is called open system."

EXAMPLE:

- Tea in a cup.
- Hot solution in a beaker.

● CLOSED SYSTEM:

"A system in which energy can be exchanged to the surrounding but matter cannot is called closed system."

EXAMPLE:

- Sealed test tube.
- Hot water in a closed iron container.

● ISOLATED SYSTEM:

"A system in which neither energy nor matter can be exchanged to the surrounding is called isolated system."

EXAMPLE:

- Tea in a thermos.
- Solution in a calorimeter.

On the basis of their physical state:**• HOMOGENOUS SYSTEM:**

"A system which has similar physical state (one phase) throughout is known as Homogenous system."

EXAMPLE:

Gas + Gas → Air

Liquid + liquid → Ester ($\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$)

Solid + Solid → Ore, alloy etc.

• HETEROGENEOUS SYSTEM:

"A system which has different physical state (two or more phase) throughout is known as Heterogeneous system."

EXAMPLE:

Gel, Cloud, and Foam etc.

"DIFFERENCE BETWEEN OPEN, CLOSE & ISOLATED SYSTEM"

Open system	Closed system	Isolated system
It is the system in which energy and matter (mass) are transferable between system and surrounding	It is the system in which only energy is transferable between system and surrounding in this system exchange of matter does not occur	It is the system is which neither energy nor matter is transferable between system and surrounding
e.g. man plant	e.g. sealed tube containing hot water	A sealed solution in thermos bottle

"SURROUNDING"**DEFINITION:**

"The environment of the system is called surrounding."

"OR"

"All the remaining portion of the universe except system is called surrounding."

"BOUNDARY"**DEFINITION:**

"The real or imaginary line which separates system from surrounding is called Boundary."

"STATE"**DEFINITION:**

"The set of macroscopic properties which describe a system completely is called state."

TYPES OF STATE:

There are two types of state.

- Initial state.
- Final state.

- **INITIAL STATE:**

"The description of the system before it suffers any change is called as the initial state of the system".

- **FINAL STATE:**

"The description of the system after it undergoes a change is known as final state of the system".

"CHANGE OR CHANGE OF STATE"**DEFINITION:**

"The difference b/w initial and final state of the system is called change."

Or

"In a thermodynamic term, if some macroscopic properties such as temperature, pressure, volume and mole are known, with the change of one or more of these properties a system changes its state is called change of state".

It is denoted by Δ .

EXAMPLE:

If T_1 is the condition of temperature at the initial state and T_2 is the condition of temperature at the final state, then

$$T_2 - T_1 = \Delta T$$

Where, ΔT is the change in temperature of the system.

“STATE FUNCTION”**DEFINITION:**

“The property of the system which depends only on initial and final state of the system and not on how it changes is called state function.”

EXAMPLE:

Enthalpy of the system is an example of state function.

“INTERNAL ENERGY”**DEFINITION:**

“The total energy possessed by molecules of the system is called internal energy”.

$$E = (\text{Rotational} + \text{vibrational} + \text{potential}) \text{ Energies}$$

“MACROSCOPIC PROPERTIES”**DEFINITION:**

“The properties of a system, rather than that of the part of the system, which are easily measurable are known as the macroscopic properties”.

EXAMPLE:

Pressure, temperature, volume etc.

“TYPES OF MACROSCOPIC PROPERTIES”

There are two types of macroscopic properties.

- INTENSIVE PROPERTIES.
- EXTENSIVE PROPERTIES.

- **INTENSIVE PROPERTIES:**

“It is a characteristic property of the system and is independent of the amount of material concerned.”

EXAMPLE:

- Density.

Pressure.
Temperature.
M.P & B.P etc.

EXTENSIVE PROPERTIES:

"It is the property that depends upon the amount of the substance present in the system."
The change in extensive property is proportional to the change in the quantity of the material in the system.

EXAMPLE:

Mass.
Volume.
Mole numbers.
Enthalpy etc.

"DIFFERENCE"

INTENSIVE PROPERTIES	EXTENSIVE PROPERTIES
➤ It is a characteristic property of the system and is independent of the amount of material concerned.	➤ It is the property that depends upon the amount of the substance present in the system.
➤ Properties which are not affected by adding or subtracting mass of substance.	➤ Properties which can be affected by adding or subtracting mass of substance.
➤ Purity of a material can be tested with the help of intensive property.	➤ Purity of a material cannot be tested with the help of extensive property.
➤ Density, temperature, pressure, M.P & B.P etc are the examples of intensive properties.	➤ Mass, volume, no of moles, enthalpy etc are the examples of extensive properties.

"ENTROPY"

DEFINITION:

"The measure of the disorder of the system is called Entropy."

EXAMPLE:

"Arrangement of particles are more disordered in liquid and gases than solids. It is directly proportional to temperature."

"ENTHALPY"

DEFINITION:

"H' called enthalpy (heat content) is introduced to express thermal changes at constant pressure."

OR

"The sum of internal energy and the product of its pressure and volume of a system is called as enthalpy."

MATHEMATICALLY:

$$\underline{H = E + PV}$$

Where,

E = Internal energy.

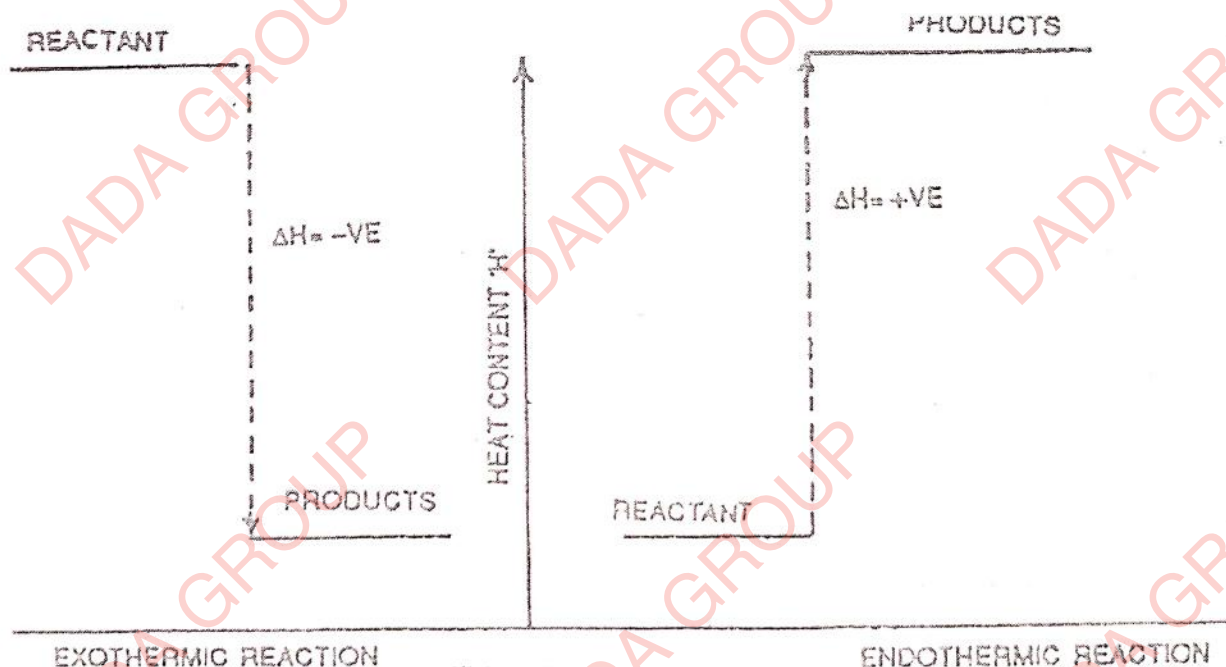
P = Pressure of the system.

V = Volume of the system.

"SIGNIFICANCE OF ΔH "**EXPLANATION:**

ΔH is the change in enthalpy. ΔH is the characteristic property of a system and depends on the initial and final state of a system and not on the means by which the changes are brought. In other words, ΔH is state function.

For all exothermic processes, ΔH would be negative and for all endothermic process, ΔH is positive.

**EXAMPLE:**

ENDOTHERMIC REACTION:



EXOTHERMIC REACTION:



"1ST LAW OF THERMODYNAMICS"

INTRODUCTION:

This law was enunciated by Helmholtz in 1847 and is also called law of conservation of energy.

STATEMENT:

"Energy can neither be created nor be destroyed, although it may change from one form to another."

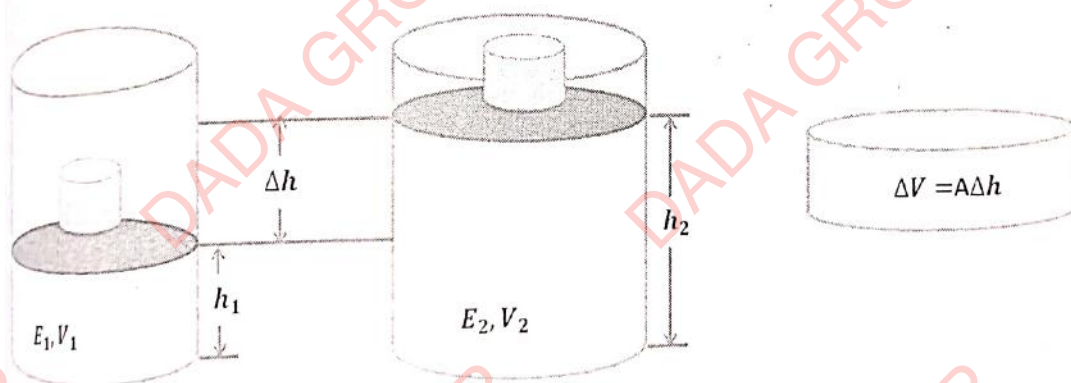
OR

"The total energy of a system and its surrounding must remain constant."

EXPLANATION:

First law of thermodynamics is one of the most fundamental of all the scientific laws. On the basis of this law, other thermodynamic laws were represented.

From the definition it is clearly understood that any gain or lose of energy by the system must be exactly equal to lose or gain of energy by the surrounding of the system. It also means that when a certain amount of energy is produced, an equivalent amount of other form of energy is used up.



MATHEMATICALLY:

Let us consider a cylinder filled with a gas having movable frictionless & weightless piston.

If the system whose internal energy is ' E_1 ', absorb heat ' q ' while ' W ' be the work done on system, the internal energy increases to ' E_2 '.

According to the law, internal energy change may be formulated as:

$$E_2 - E_1 = q + W$$

$$\Delta E = q + W$$

“SIGN CONVENTION”

- If heat is absorbed by the system, 'q' will be positive.
- If heat is given out by the system, 'q' will be negative.
- If work is done on the system by the surrounding then 'w' will be positive.
- If work is done by the system on the surrounding then 'w' will be negative.

“PRESSURE VOLUME WORK EQUATION”**DEFINATION:**

“The work which is done by expansion or compression of a gas in a thermodynamic system is known as pressure volume work.”

CONSIDERATION:

Consider a gas present in a cylinder having a weightless frictionless movable piston.

Let,

Area of cross section of piston = A

Pressure on the piston = P

Initial height of the piston = h_1

Final height of the piston = h_2

Distance covered by piston = Δh

MATHEMATICALLY:

As we know that, the pressure is defined as the force per unit area. Therefore,

$$P = \frac{F}{A}$$

So,

$$F = P.A \rightarrow (1)$$

If the force is exerted by the gas is greater than external pressure then piston moves upward and does some work:

$$W = - F.S \rightarrow (2)$$

Since a/c to equation (1),

$$F = PA$$

And,

$$S = \Delta h$$

So equation (2) become,

$$W = - P \Delta. \Delta h$$

Since,

$$A = l \times b$$

Where,

l = length

b = breadth

So we can write as,

$$W = - P (l \times b) \Delta h$$

Since,

$$V = l \times b \times h$$

Where,

l = length.

b = breadth.

h = height.

So,

$$W = - P \Delta V \rightarrow (3)$$

According to the 1st law of thermodynamics,

$$\Delta E = q + W$$

Using equation (3) in above,

$$q = \Delta E + P \Delta V$$

This equation is known as pressure volume work equation.

APPLICATION OF 1ST LAW OF THERMODYNAMICS

WORK AT CONSTANT VOLUME:

(ISOCORIC PROCESS)

Let 'q' be the heat which is absorb by the system at constant volume.

MATHEMATICALLY:

We know that,

$$q = \Delta E + P\Delta V$$

But, at constant volume,

$$\Delta V = 0$$

Therefore,

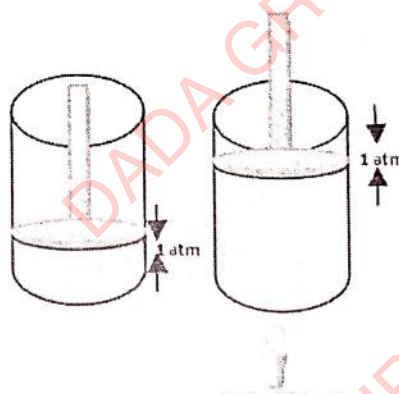
$$q_v = \Delta E + P(0)$$

$$q_v = \Delta E$$

Thus, at constant volume heat is absorb or released by the system is equal to energy change of the system.

WORK AT CONSTANT PRESSURE:**(ISOBARIC PROCESS)**

Let 'q' be the heat which absorbed by the system at constant pressure. Due to this heat energy, the internal energy of the system E_1 becomes E_2 and the volume changes from V_1 to V_2 .

**MATHEMATICALLY:**

We know that,

$$\Delta E = q - P\Delta V$$

$$q_p = E_2 - E_1 + P(V_2 - V_1)$$

$$q_p = E_2 - E_1 + PV_2 - PV_1$$

$$q_p = E_2 + PV_2 - (E_1 + PV_1) \rightarrow (1)$$

since,

$$H_2 = E_2 + PV_2$$

$$H_1 = E_1 + PV_1$$

Putting in equation (1),

$$q_p = H_2 - H_1$$

$$q_p = \Delta H$$

Hence given equation can be written as,

$$\Delta H = \Delta E + P\Delta V$$

The amount of heat absorbed or evolved at constant pressure becomes equal to enthalpy change.

"HESS'S LAW"

INTRODUCTION:

In 1840, G. H. HESS discovered a very important generalization which is known as Hess's law.

STATEMENT:

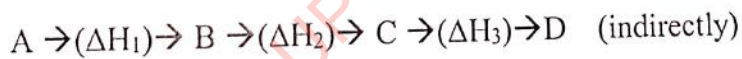
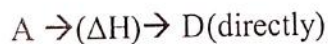
"The heat change (enthalpy) for a chemical reaction is the same whether it takes place in several steps or directly in one step."

EXPLANATION:

From Hess's law it is clear that the enthalpy change for a chemical reaction in one step directly is equal to the sum of all enthalpies of indirect steps as shown below,

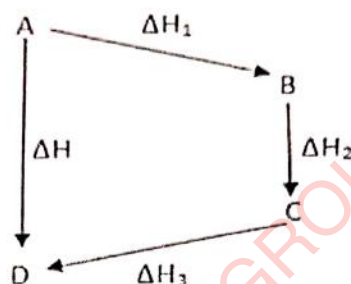
$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$

EXAMPLE:

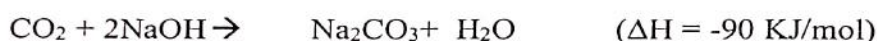


According to the law,

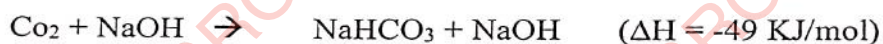
$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$

**VERIFICATION OF HESS'S LAW:****1ST METHOD:**

CO₂ gas reacts with excess of NaOH solution and form sodium bicarbonate and water. For this reaction the enthalpy change is -90 KJ/mol. This reaction may take place in two steps. As shown below,

**2ND METHOD:**

For first step, formation of NaHCO₃ with the enthalpy change is -49 KJ/mol. As shown below,



For second step, formation of Na₂CO₃ with the enthalpy change is -41 KJ/mol. As shown below,



According to the law,

$$\Delta H = \Delta H_1 + \Delta H_2$$

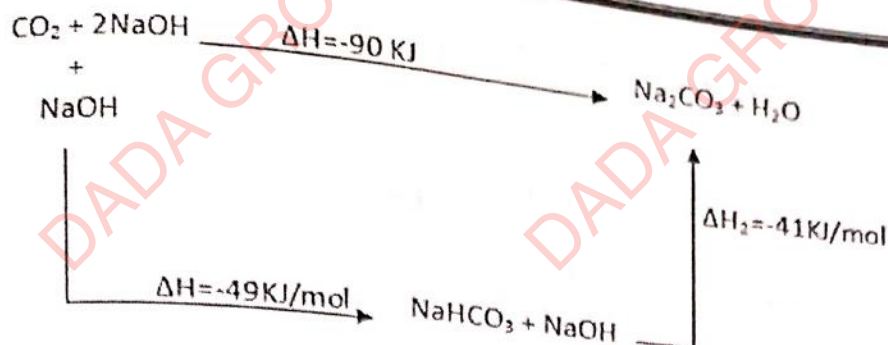
$$\Delta H = (-49-41) \text{ KJ/mol}$$

$$\Delta H = -90 \text{ KJ/mol}$$

Hence,

The enthalpy change in direct step -90 KJ/mol is equal to the sum of all enthalpies of indirect steps (-49-41) kJ/mol.

Thus, the law is verified.



APPLICATIONS OF HESS'S LAW:

- With the help of Hess's law we can calculate the heat of formation of different compounds.
- Lattice energy of several ionic compounds can be calculated.
- Thermo chemical equations can be treated as algebraic equations.

"HEAT OF FORMATION"

DEFINITION:

"The enthalpy change when one mole of any compound is formed from its elements is called heat of formation."

REPRESENTATION:

is represented by ΔH_f .

SAMPLE:



"STANDARD HEAT OF FORMATION"

DEFINITION:

The heat of formation at 25° C and 1 atm pressure is known as standard heat of formation.

REPRESENTATION:

is represented by ΔH_f° .

EXPLANATION:

Conventionally ΔH_f° , of most of the elements (H_2 , O_2 , N_2 , Na, etc) are taken as zero because no further change is needed to bring them to standard state condition. However, for those elements which exists in more than one allotropic forms, the most stable allotropic form is conventionally assumed to have $\Delta H_f^\circ = 0$.

SAMPLE:

Graphite, O_2 is more stable than Ozone, that's why ΔH_f° of O_2 is taken as zero.

"STANDARD ENTHALPY OF REACTION"**DEFINITION:**

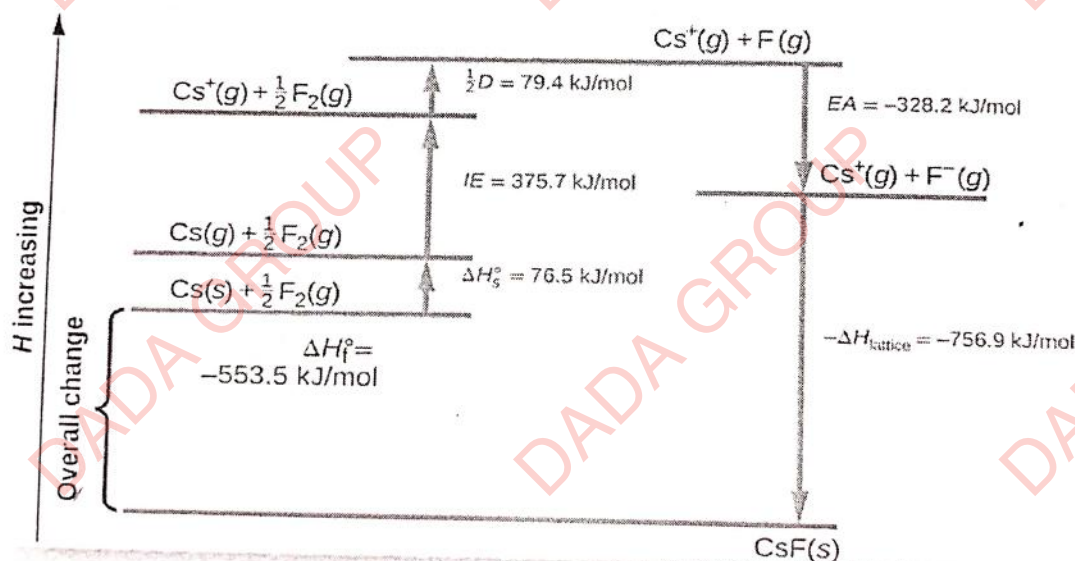
It is the enthalpy absorbed or evolved when fixed number of moles of reactants represented in a balance chemical equation are completely reacted to form product with the condition that all reactants and products are in their standard states.

FORMULA:

$$\Delta H_{\text{REACTION}}^{\circ} = \sum np \Delta H_f^{\circ} (\text{products}) - \sum nr \Delta H_f^{\circ} (\text{reactants})$$

"BORN HABER CYCLE"**INTRODUCTION:**

Max Born and Fritz Haber (1919) to correlate the lattice energy of ionic solids with some other energies involved in their formation.

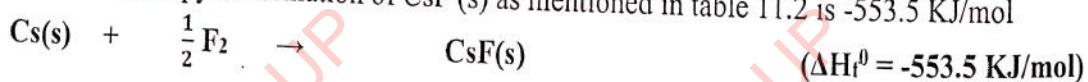
ILLUSTRATION:**STATEMENT:**

"It is a cycle of enthalpy change in which an ionic solid is theoretically formed from its elements in their standard states".

EXPLANATION:

To understand, how the lattice energy of a binary ionic solid determines by using Born Haber cycle let's take solid Cesium fluoride (CsF) as an example. We start by writing a thermochemical equation for the formation of CsF from its elements at their standard states (25° C and 1 atm).

The standard enthalpy of formation of CsF (s) as mentioned in table 11.2 is -553.5 KJ/mol



Various energies involved in the formation of CsF(s) are described in the following stepwise

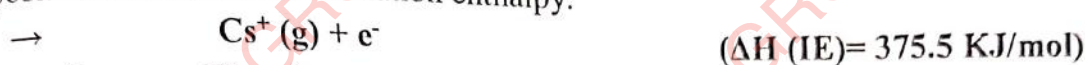
Step 1: Sublimation of Cesium:

Involves the vaporization of metallic cesium into gaseous atom. The process is endothermic because heat is always needed to sublime a solid.



Step 2: Ionization of Cesium atom:

Involves the loss of electron from cesium gaseous atom. The enthalpy change corresponds to this process is always positive and known as ionization enthalpy.



Step 3: Dissociation of gaseous Fluorine:

Involves the breaking of F_2 molecule into fluorine gas atoms. The enthalpy related to this process is always positive and labeled as enthalpy of dissociation or often known as enthalpy of atomization.



Step 4: Formation of F^- ion:

Involves the absorption of electron into fluorine gas atom. The enthalpy change of this thermochemical process is negative and specify as electron affinity.



Step 5: Formation of Solid CsF :

Involves the electrostatic interaction of oppositely charged gaseous ions. The enthalpy change in this process is indeed negative and known as Lattice energy.



According to Hess's Law, the sum of enthalpy change of these five steps will be equal to the enthalpy of formation of CsF(s) being the overall enthalpy change of the energy cycle is zero. This may be set formulated as:

$$\Delta H_f^\circ = \Delta H_{\text{sub}} + \Delta H (\text{I.E.}) + \frac{1}{2} \Delta H (\text{D}) + \Delta H (\text{E.A.}) + \Delta H (\text{L.E.})$$

Substituting the values of concerned energies, we get lattice energy.

$$(-553.5) = (76.5) + (375.5) + (\frac{1}{2} \times 157) + (-328.2) + (\Delta H (\text{LE}))$$

$$\Delta H (\text{LE}) = -755.8 \text{ KJ/mol}$$

"OBJECTIVES"

FILL IN THE BLANKS:

- The branch of Chemistry, which deals with the heat changes that take place during chemical reaction, is called _____.
- The branch of science which deals with energy changes accompanying physical and chemical transformation is called _____.

Heat absorbed in a chemical reaction is called _____.

4. Such reactions in which heat is evolved are called _____ reactions.
5. Such reactions in which heat is absorbed are called _____ reactions.
6. In exothermic reactions, heat evolved is given by _____ sign of ΔH .
7. In endothermic reactions heat absorbed is given by _____ sign of ΔH .
8. The total heat change in a reaction is the same whether it takes place in one or several steps.
9. The first law of thermodynamics is also known as _____.
10. The part of universe under observation is called _____.
11. Heat change at constant pressure from initial to final state of the system is simply equal to the _____.
12. Such properties, which give description of a system at a particular moment, are called _____.
13. The term $E + PV$ is called _____.
14. ΔH represents change in _____.
15. Heat absorbed by the system at constant volume is completely utilize to increase the _____ of the system.
16. The characteristic properties of a system which is independent of amount of material concerned is called _____ properties.
17. The characteristic properties of a system which depend on amount of substance present in it are called _____ properties.
18. Density, pressure and temperature are the examples of _____ properties.
19. Mole numbers and enthalpy are the examples of _____ properties.
20. A system, which exchange both energy and matter with its surrounding, is called _____ system.
21. A system, which only exchange energy with the surrounding but not matter is, called _____ system.
22. A system which neither exchange energy nor matter with its surrounding is called _____ system.
23. A system is _____ if it contains only one phase.
24. A system is _____ if it contains more than one phase.
25. Hess's Law is employed to calculate _____ of a chemical reaction

Standard enthalpies are measured at _____.

First law of Thermodynamics is mathematically represented as _____.

The work done (w) is mathematically denoted by _____.

The change in enthalpy is denoted by _____.

_____ law is used in calculating heat of reaction.

_____ is defined as the change in enthalpy when one gram mole of a compound is produced from its element.

Heat of formation is denoted by _____.

When the work is done on the system by the surrounding the sign of work done (w) is _____.

When the work is done by the system on surrounding the sign of work done is _____.

MULTIPLE CHOICE QUESTIONS

The quantity of heat evolved or absorbed during a chemical reaction is called _____.

(Heat of Reaction, Heat of Formation, Heat of Combination)

An endothermic reaction is one, which occurs _____.

(With evolution of heat, with absorption of Heat)

An exothermic reaction is one during which _____.

(Heat is liberated, Heat is absorbed, no change of heat occurs)

Thermo-chemistry deals with _____.

(Thermal Chemistry, Mechanical Energy, Potential Energy)

Enthalpy is _____.

(Heat content, Internal energy, Potential Energy)

Hess's Law is also known as _____.

(Law of conservation of Mass, Law of conservation of Energy, Law of Mass Action)

Anything under examination in the Laboratory is called _____.

8. The factor $E + PV$ is known as _____.
(Heat content, Change in Enthalpy, Work done)
9. The enthalpy change when a reaction is completed in single step will be _____ as compared to that when it is completed in more than one steps.
(Equal to, Partially different from, Entirely different from)
10. The heat absorbed by the system at constant _____ is completely utilized to increase the internal energy of the system.
(Volume, Pressure, Temperature)
11. _____ property of a system is independent of the amount of material concerned.
(Intensive, Extensive, Physical)
12. _____ property of a system depends upon the amount of substance present in the system.
(Intensive, Extensive, Physical)
13. _____ is defined as the change in enthalpy when one gram mole of a compound is produced from its elements.
(Heat of Reaction, heat of Formation, Heat of Neutralization)

TRUE AND FALSE

1. The SI unit for measuring heat change is kilocalories.
2. Heat is evolved in exothermic reaction.
3. Heat is evolved in exothermic reaction is given by positive sign of ΔH .
4. Heat is absorbed in endothermic reaction.
5. Heat absorbed in endothermic reaction is given by negative sign of ΔH .
6. Burning of natural gas is an endothermic reaction.
7. A substance having high heat of formation may decompose easily.
8. Hess's Law states that energy can neither be created nor destroyed.
9. First Law of thermodynamics is also known as Law of conservation of Energy.
10. $E + PV$ is called Heat content.
11. Heat content is also known as Enthalpy.

Heat of formation of a compound is defined as the change in enthalpy when one gram of a compound is defined as the change in enthalpy when one gram of a compound is produced from its element.

A positive sign is given to Heat of Formation of compounds produced in endothermic reaction.

A system which contains only one phase is called a heterogeneous system.

Intensive property depends upon amount of substance present in the system.

Extensive property of a system is independent of amount of material concerned.

Any material or collection of material under examination in the laboratory is called a system.

An open system is that in which only exchange of energy with the surrounding is possible.

If the work is done on a system by surrounding 'w' is positive.

The resultant heat change in a chemical reaction is the same whether the reaction takes place in one or several steps.

The energy supplied to a system at constant volume is used in changing the internal energy of the system.

The energy supplied to a system at constant pressure is used in changing the internal energy of the system.

Thermochemistry is the branch of chemistry, which deals with electrical energy.

Standard enthalpy of formation of a substance is always negative.

Standard enthalpies are measured at 298°K .

1 Calorie = 4184 KJ.

Thermodynamics can also be defined as the study of flow of heat.

The total energy of the system and its surroundings remains unchanged.

No heat is evolved when CH_4 is burnt.

Heat evolved in a chemical reaction may be utilized to perform mechanical work.

"NUMERICALS"

LAW OF THERMODYNAMICS:

1. When 5400 joules of heat is added to a system of gas at constant pressure its internal energy increases by 1000 joules. Calculate the change in volume of the system.

0.0434m³

2. A system absorbs 2000 joule of heat from the surrounding and does 1200 J of work on the surrounding by expansion. Find the internal energy change of the system. **800 J**
3. A system absorbs 200 J of heat from the surrounding and does 120 J of work on the surrounding by expansion. Find the internal energy change of the system **80 J**
4. 600 J of heat was given to a gas at $1.01 \times 10^5 \text{ N/m}^2$ pressure. The internal energy was increased by 300 J. Find the increase in volume. **$2.97 \times 10^{-3} \text{ m}^3$**
5. 800 J of work is done on the system which gives 500 J of heat. What is the value of ΔE for the process? **1300 J**
6. If 1500 calories of heat are added to a system while system does the work equivalent to 2500 calories by expanding against the surrounding, find the value of ΔE for the system. **-158.5 J**
7. In a certain process 848 J of heat is absorbed by a system, while 394 J of work is done on the system. What is the change in internal energy of the process? **1242 J**
8. 3600 KJ of heat is entered in a gas cylinder. Calculate the value of ΔE . If volume is kept constant. **3600 KJ**
9. If volume is not constant, work of -800 kJ was performed by the gas. **4400 KJ**
10. If the gas is allowed to expand, the value of work is 5200KJ. **-1600 KJ**
11. In a certain process 700 J of heat is absorbed by a system, while 300 J of work is done on the system. What is the change in internal energy of the process? **1000J**
12. In a certain process 500 J of work is done on the system which gives of 200 J of heat. What is the value of ΔE for the process? **300 J**

HEAT OF FORMATION:

1. Calculate the heat of formation from the following data:

$2\text{Na} + \frac{1}{2} \text{O}_2$	$\rightarrow$	Na_2O	$\Delta H = ?$
$\text{Na} + \text{H}_2\text{O}$	$\rightarrow$	$\text{NaOH} + \frac{1}{2} \text{H}_2$	$\Delta H = -43.2 \text{ Kcal/mole}$
$\text{Na}_2\text{O} + \text{H}_2\text{O}$	$\rightarrow$	2NaOH	$\Delta H = -63.2 \text{ Kcal/mole}$
$\text{H}_2 + \frac{1}{2} \text{O}_2$	$\rightarrow$	H_2O	$\Delta H = -68.0 \text{ Kcal/mole}$

91.2 Kcal/mol
2. Calculate ΔH for the reaction:

$3\text{Mg} + \text{N}_2$	$\rightarrow$	Mg_3N_2	$\Delta H = ?$
$3\text{Mg} + 2\text{NH}_3$	$\rightarrow$	$\text{Mg}_3\text{N}_2 + 2\text{NH}_3$	$\Delta H = -371 \text{ KJ}$
$\frac{1}{2}\text{N}_2 + \frac{3}{2} \text{H}_2$	$\rightarrow$	NH_3	$\Delta H = -46 \text{ KJ}$

-436 KJ
3. Calculate the standard heat of formation of the methyl alcohol from its element from the following data:

$\text{C} + 2\text{H}_2 + \frac{1}{2} \text{O}_2$	$\rightarrow$	CH_3OH	$\Delta H_f = ?$
$\text{C} + \text{O}_2$	$\rightarrow$	CO_2	$\Delta H = -394 \text{ KJ/mole}$
$\text{H}_2 + \frac{1}{2} \text{O}_2$	$\rightarrow$	H_2O	$\Delta H = -286 \text{ KJ/mole}$
$\text{CH}_3\text{OH} + \frac{3}{2} \text{O}_2$	$\rightarrow$	$\text{CO}_2 + 2\text{H}_2\text{O}$	$\Delta H = -726 \text{ KJ/mole}$

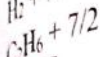
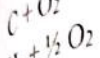
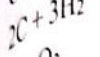
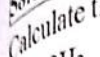
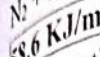
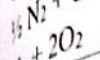
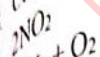
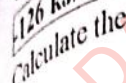
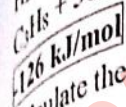
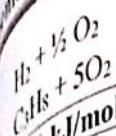
-240 kJ/mol
4. Calculate the heat of formation of propane from the following data:

$3\text{C} + 4\text{H}_2$	$\rightarrow$	C_3H_8	$\Delta H = ?$
$\text{C} + \text{O}_2$	$\rightarrow$	CO_2	$\Delta H = -394 \text{ kJ/mole}$

THERMOCHEMISTRY

Chapter# 11

Chemistry I



$$\Delta H_f^\circ = -286 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -2200 \text{ kJ/mole}$$

Calculate the heat of formation of N_2O_4 from the following data:

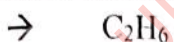


$$\Delta H_f^\circ = ?$$

$$\Delta H_f^\circ = +33.95 \text{ kJ/mole}$$

$$\Delta H_f^\circ = +9.3 \text{ kJ/mole}$$

Calculate the heat of formation of ethane at 25°C from the following data:



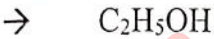
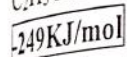
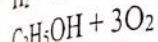
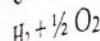
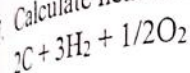
$$\Delta H_f^\circ = ?$$

$$\Delta H_f^\circ = -394 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -286 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -1560 \text{ kJ/mole}$$

7. Calculate heat of formation for the reaction:



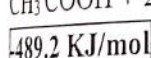
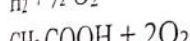
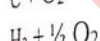
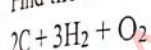
$$\Delta H_f^\circ = ?$$

$$\Delta H_f^\circ = -393.5 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -286 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -1369 \text{ kJ/mole}$$

8. Find the heat of formation of Acetic acid at 25°C :



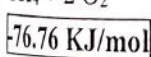
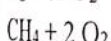
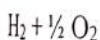
$$\Delta H_f^\circ = ?$$

$$\Delta H_f^\circ = -394 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -286 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -870.8 \text{ kJ/mole}$$

9. Determine ΔH for the reaction from following data:



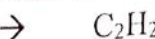
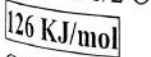
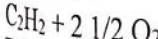
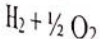
$$\Delta H_f^\circ = ?$$

$$\Delta H_f^\circ = -394 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -286 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -890.34 \text{ kJ/mole}$$

10. Calculate heat of formation of acetylene (C_2H_2):



$$\Delta H_f^\circ = ?$$

$$\Delta H_f^\circ = -394 \text{ kJ/mole}$$

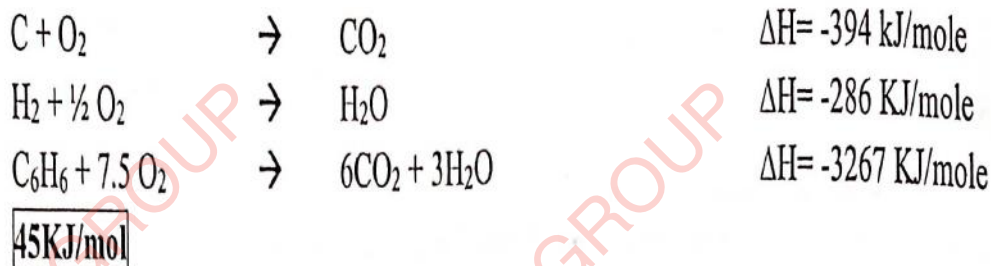
$$\Delta H_f^\circ = -286 \text{ kJ/mole}$$

$$\Delta H_f^\circ = -1200 \text{ kJ/mole}$$

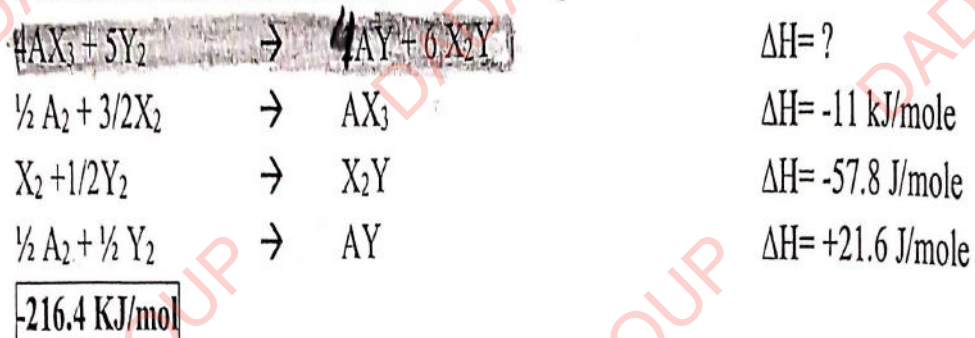
11. Calculate heat of formation of benzene (C_6H_6) from following data:



$$\Delta H_f^\circ = ?$$



12. Calculate the heat of formation from the following data:



-57800
+21600

DADA OFFICIAL

$\Delta H = 1000 \text{ KJ}$

$-11J =$

CHAPTER - 12

ELECTROCHEMISTRY

DEFINITION:

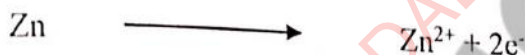
Electro chemistry deals with the chemical change happen due to electric current as well as the generation of electric current by chemical change. It is best defined as,
"The study of interchange of chemical and electrical energy".

"OXIDATION"

DEFINITION:

"An increase in the oxidation number of an element or ion during a chemical change by the loss of electron in a chemical reaction is called oxidation."

EXAMPLE



"REDUCTION"

DEFINITION:

"A decrease in the oxidation number of an element or ion during a chemical change by the gain of electron in chemical reaction is called reduction."

EXAMPLE



"OXIDATION NUMBER"

DEFINITION:

The apparent positive or negative charge which an atom would have in the compound is known as oxidation number".

Using the concept of oxidation number, all the compounds are imagined to be completely ionic and term formal charge means, not the actual charge of an atom in the molecule or ion.

The determination of oxidation number of an atom is followed by following general rules.

- The oxidation number of any element in the Free State is zero for e.g. H_2 , Cl_2 , N_2 , Br_2 and O_2 have zero oxidation number
- The oxidation number of hydrogen in compound is $1+$ except in hydride such as NaH where it is $1-$
- The oxidation number of oxygen in compound is $2-$ except in peroxide
- The oxidation number of each element of group VIIA, halogen in binary compound is $1-$
- The oxidation number of each element of group IA, IIA and IIIA is $1+$, $2+$ and $3+$
- The algebraic sum of oxidation number of all atoms in the formula of a neutral compound equal zero
- The oxidation number of any ion is equal to the charge on the ion

"OXIDIZING AGENTS"**DEFINITION:**

"An atom or ion which picks up electron and show decrease in oxidation number is called an oxidizing agent or those which easily reduce (gain of electron) and oxidized other is known as oxidizing agent."

EXAMPLE:

Here HNO_3 is oxidizing agent

"REDUCING AGENTS"**DEFINITION:**

"An atom or ion which loss electron and show increase in oxidation number is called on reducing agent."

EXAMPLE:

Here H_2S is reducing agent.

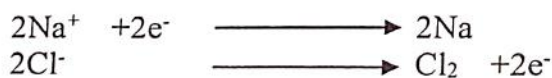
"OXIDATION REDUCTION REACTIONS"**DEFINITION:**

"Such reaction which is brought about by loss of electrons (oxidation) and gain of electron (reduction) simultaneously is called an Oxidation Reduction reaction or Redox reaction."

The Redox reaction can be considered as made up of two of reaction



Here,

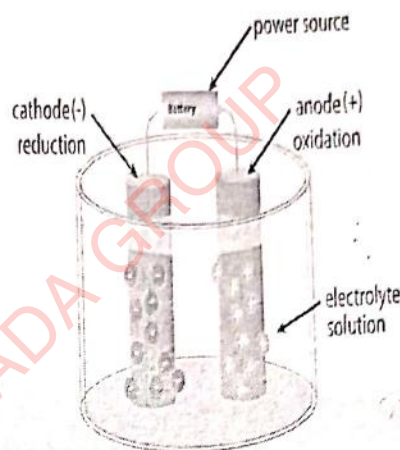
**Characteristics of Redox Reactions:**

There are some general characteristics of redox reactions.

- All redox reactions may involve the transfer of electrons.
- Oxidation and reduction process always occur side by side thus if one substance is oxidized the other is reduced at the same time.
- Not all atoms in a redox reaction need to change their oxidation states but at least one atom should be oxidized and one should be reduced.

"ELECTROLYTIC CELL"

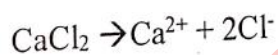
ELECTROLYTIC CELL PARTS



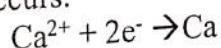
An electrolytic cell is an apparatus containing two electrodes (rods or foils) known as electrodes dipped in an electrolytic solution and connected to a battery wherein an electric current is used to derive a non-spontaneous reaction. The electrode connected to the negative terminal of battery is called cathode, through which electrons enter the solution whereas the electrode connected to the positive terminal of battery is called anode, through which electrons leave the cell. The electrodes are basically metallic plates, rods or foils through which the current enters or leaves the electrolyte. At the negative electrode (Reduction occurs) and at the positive electrode (Oxidation occurs). The electrolyte spontaneously dissociates to produce ions within the solution. The positive ions are known as cations while the negative ions are known as anions.

When the electric current is passed through the solution, anions move towards anode where they lose electrons to get oxidized. On the other hand, cations move towards cathode where they gain electrons and become reduced. By this we can say that oxidation occurs at anode while reduction occurs at cathode.

Electrolysis of Calcium Chloride.



Reaction at cathode at which, reduction occurs:



Reaction at anode at which oxidation occurs:



"ELECTROLYSIS"

DEFINITION:

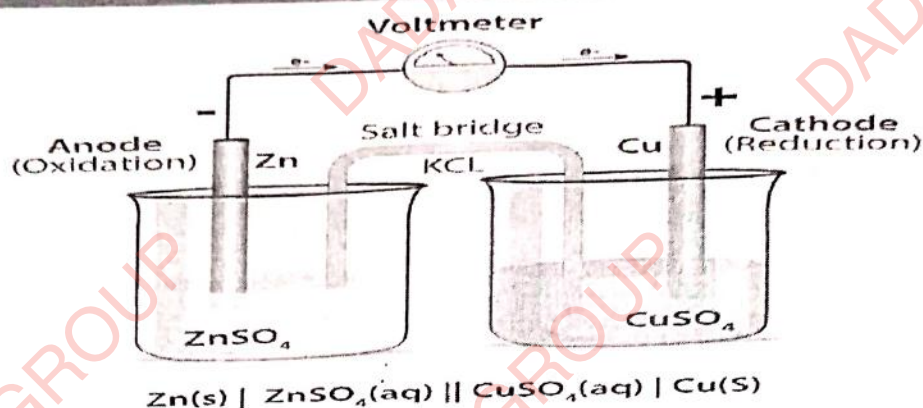
"The movement of cation and anion towards their respective electrodes accompanying all chemical change in the electrolytic solution under the influence of electric current is known as electrolysis".

"GALVANIC CELL"

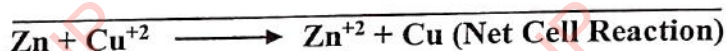
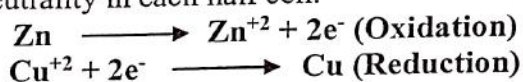
A Galvanic or voltaic cell is type of electrochemical cell in which current is generated by a spontaneous redox reaction. Galvanic cell is typically consist of two half cells which are interconnected electrically. Each half cell is made up of a strip of metal immersed into a solution of its metal ions.

VARIATION OF CELL POTENTIAL IN ZN-CU CELL

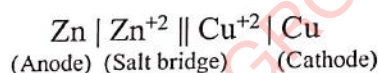
BYJU'S



"A salt bridge is a u-shaped glass tube in which an inert electrolyte is filled along with gel. It allows the flow of ions but prevent the mixing of solution".
Thus maintain the electrical neutrality in each half cell.



Galvanic cells are conveniently express in short notation,



Single vertical lines show phase boundary and double vertical line shows salt bridge.

"CELL POTENTIAL"

The electrical energy produce by voltaic cell is due to the electron flow from anode to cathode through an external wire.

"The driving force with which the electrons pushes out from anode into external circuit is known as electromotive force (emf) or cell potential or cell voltage".

The standard electrode potential of a single half cell is defined as,

"The difference of potential created between metal electrode and 1 molar solution of its ions at 25°C and 1 atmospheric pressure". It is abbreviated by E° and measured in volt.

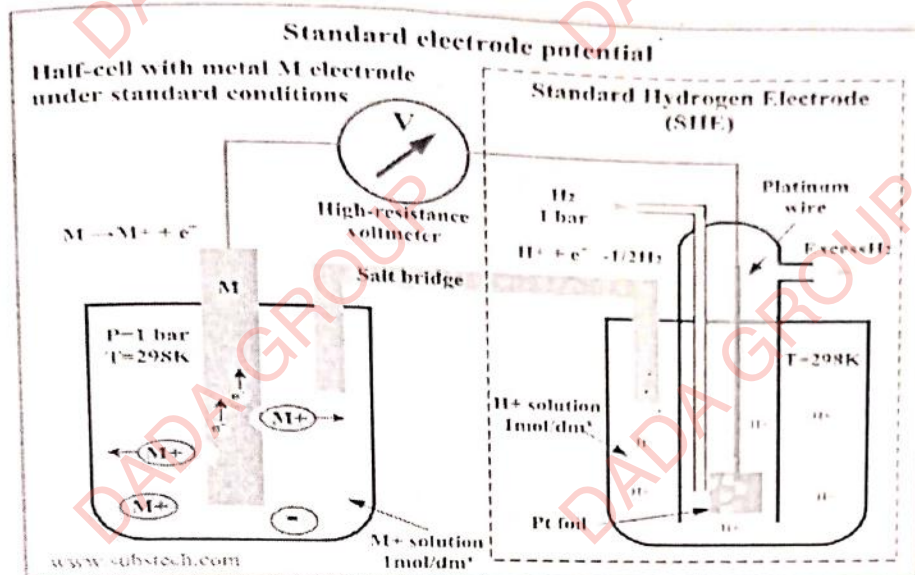
"ELECTRODE POTENTIAL"

DEFINITION:

"The potential difference between electrode and the solution in which it is dipped is known as ELECTRODE POTENTIAL".

OR

The potential difference created between a metal and solution of its salt is called **ELECTRODE POTENTIAL**



OXIDATION AND REDUCTION POTENTIAL:

Oxidation potential: The potential of an electrode at which oxidation takes place is called oxidation potential.

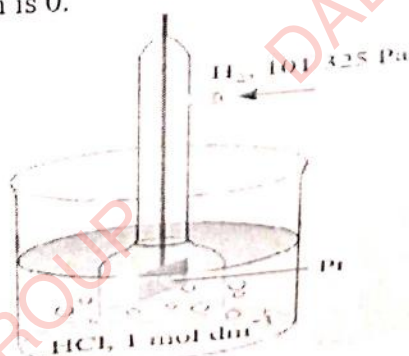
Reduction potential: The potential of an electrode at which reduction takes place is called reduction potential.

CONSTRUCTION OF STANDARD HYDROGEN ELECTRODE (SHE):

The hydrogen electrode at standard conditions of temperature, pressure and the concentrations of the solution is called a standard hydrogen electrode and the potential difference is known as standard electrode potential.

A hydrogen electrode consists of platinum plate immersed in 1M solution HCL at 25°C. A current of pure hydrogen gas is passed through the solution under the pressure of 1 atmosphere. The platinum adsorbs hydrogen gas on its surface and the platinum coated with hydrogen behaves as hydrogen electrode.

The electrode potential of standard electrode is assigned a potential of 0.00(ZERO) volts. Therefore the electrode potential of hydrogen is 0.

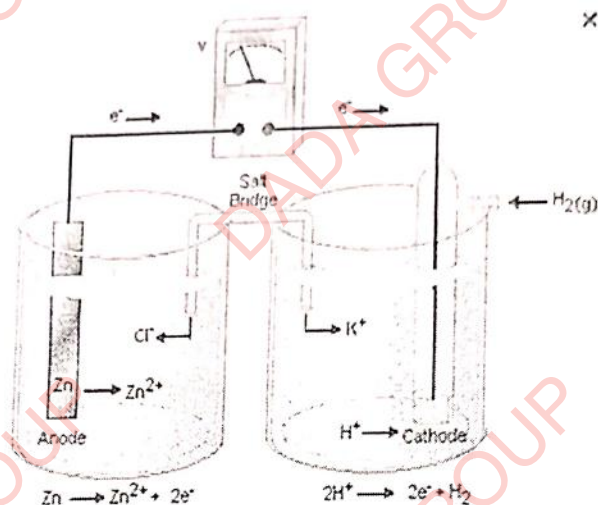


DETERMINATION OF ELECTRODE POTENTIAL:

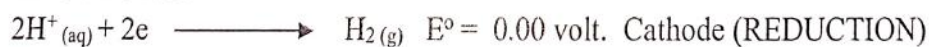
The absolute electrode potential of a single electrode cannot be determined. However the electrode potential of an electrode can be determined by comparing it with hydrogen electrode.

"ELECTRODE POTENTIAL OF ZINC ELECTRODE"**CONSTRUCTION:**

To determine the electrode potential of zinc we have to construct a voltaic cell in which zinc electrode is dipped in 1M solution of ZnSO_4 and is connected with hydrogen electrode. Salt bridge is made of KCl jelly to complete the circuit between half-cell and prevents mixing of solution.



The potentiometric reading gives the electromotive force of the cell to be 0.76 volts. Since the hydrogen electrode has the potential of 0.00 volt, Hence 0.76 volts is the potential of zinc electrode. For the external circuit it is observed that oxidation takes place at zinc and reduction takes place at hydrogen. Therefore

CELL REACTIONS:

$$E^\circ_{\text{Zn} / \text{Zn}^{+2} (1\text{M})} = + 0.76 \text{ volt}$$

Or

$$E^\circ = + 0.76 \text{ volt}$$

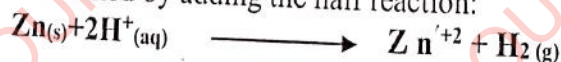
Therefore,

Oxidation potential of Zn = + 0.76 volt.



Hence the standard reduction potential of zinc is - 0.76 volt.

The cell reaction is, therefore obtained by adding the half reaction:

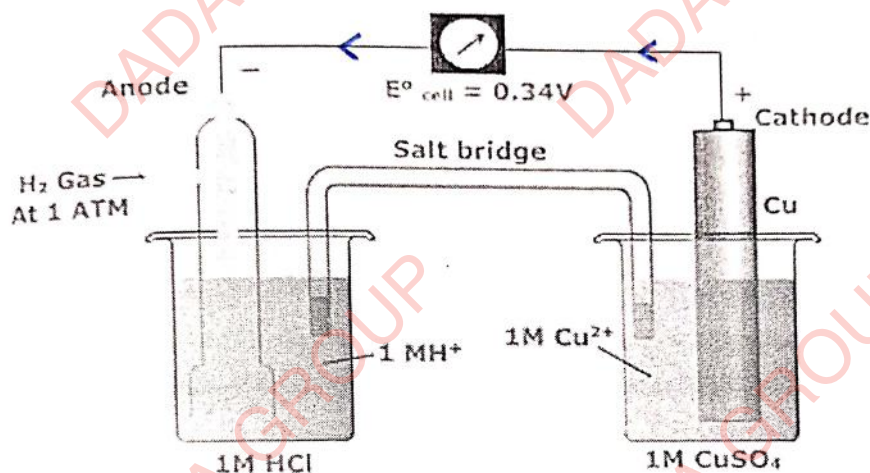


"ELECTRODE POTENTIAL OF COPPER ELECTRODE"

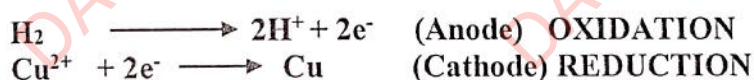
Electrode potential of can also determine by Hydrogen as reference electrode. Voltaic cell which constructed for this purpose consists of copper and hydrogen electrode.

INSTRUCTION:

first half cell consists of a strip of copper immersed in "1 M" of CuSO_4 solution while the other half cell consists of platinum foil immersed in 1M of H_2SO_4 solution and H_2 gas passes through the solution of 1 atmospheric pressure. Two half cells connected through KCl salt bridge and the two electrode join through potentiometer. In the external circuit it is observed that the direction of flow of electrons from hydrogen to copper electrode therefore hydrogen act as anode and copper as cathode. Therefore at hydrogen oxidation takes place.



Cell reactions:



Emf of cell:

The emf cell as noted by potentiometer is 0.34V.

Therefore

$$\begin{aligned} \text{Emf} &= E_o(\text{anode}) + E_R(\text{cathode}) \\ 0.36 &= 0.0 + E_R(\text{cathode}) \\ 0.34\text{V} &= E_R(\text{copper}) \end{aligned}$$

-0.34 = 0.0 - E°(anode)
E°(anode) = 0.34

"ELECTROCHEMICAL SERIES"

This is an arrangement of elements or ions in the decreasing order of their standard reduction potential values is called electrochemical series

The important facts of electrochemical series are given below.

- i) The electrode potential of hydrogen is zero and it is taken as reference electrode
- ii) Metal above hydrogen in ECS undergo oxidation and act as anode in cell
- iii) Metal below hydrogen in ECS undergo reduction and act as cathode. The order of reducing strength decrease and oxidizing strength increase from top to bottom
- iv) All metal above hydrogen -ve electrode potential below hydrogen +ve electrode potential.

"BATTERIES"

"A battery consists of two or more Galvanic cells connected in series and is used to convert chemical energy into electrical energy by means of reduction-oxidation (redox) reactions".

Have you ever seen battery in your automobile car, or the devices like cellular phone, radio, torch, toys etc? You might note that these are light weight, compact in size and work at constant voltage, a commercial battery should fulfill all these criteria. Batteries are classified into following two types.

- Primary Cells/Battery
- Secondary Cell/Battery
- Fuel Cell

PRIMARY CELL:

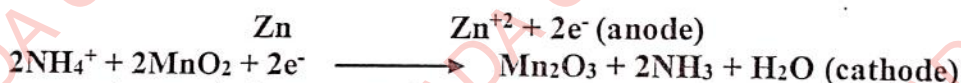
In this type of batteries redox reaction takes place in only one direction and cannot be reversed. Thus these batteries cannot be recharged or reused. When all amount of electrolyte present in the cell is consumed, they become dead.

The most inexpensive and commonly used primary cell is dry cell which we often use in remote control, transistors, flash lights and many other domestic works.

> DRY CELL:

The common dry cell was first invented by a French chemist Leclanche. The outer body of this cell is made up of zinc which serves as anode where as a graphite rod located at the centre works as cathode. The whole zinc container is lined with porous paper which works as a separator. The region between cathode and anode is filled with a paste of MnO_2 , Carbon, NH_4Cl and ZnCl_2 .

When the cell starts working Zn anode oxidizes to Zn^{+2} ions whereas at graphite cathode Mn^{+4} ions reduces to Mn^{+3} ions.



The electrons generated at anode flow out to the external circuit. This cell produces 1.5 volt electric current and become dead after complete utilizing zinc electrode. O

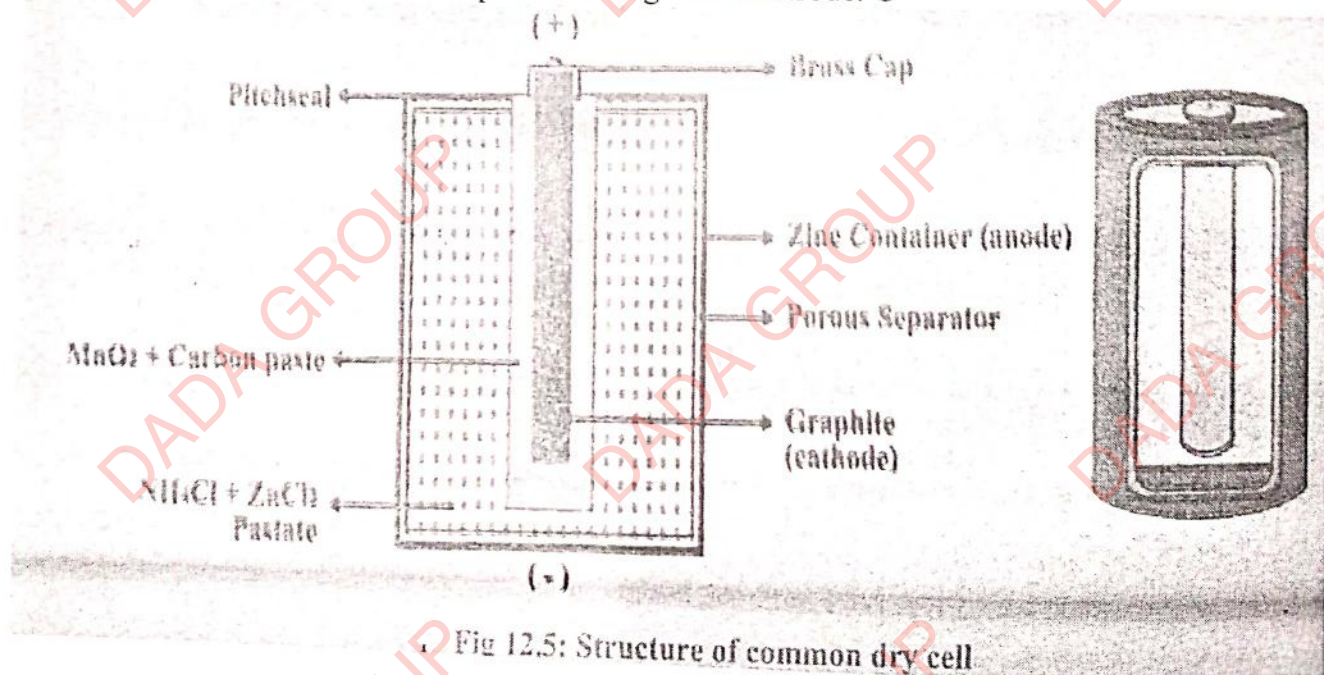
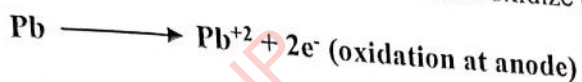


Fig 12.5: Structure of common dry cell

SECONDARY CELL:

These batteries can be recharged by reversing the electricity flow through the cell by utilizing a source of external power supply after every use. Lead storage battery which is commonly known as car battery is an example of secondary battery.

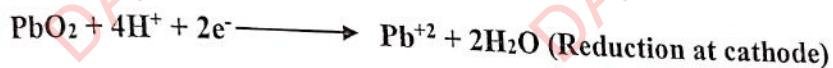
A 12 volt car battery is made up of six Galvanic cells connected in a series each of which has an emf of 2 volt. All anode plates are made up of lead (Pb) while all cathode plates consists of lead coated with PbO_2 . Both groups of plates are alternatively suspended into a dilute solution of H_2SO_4 of specific gravity 1.25. When the battery starts working, lead atoms at anode oxidize to form lead ions.



The Pb^{+2} ions combine with SO_4^{-2} ions to form solid PbSO_4 which is deposited on anode.



The electrons generate at anode pushes out into external circuit and then re-enter into cell from cathode (PbO_2) where PbO_2 reduces to Pb^{+2} ions.



The Pb^{+2} ions again combine with SO_4^{-2} ions of electrolyte and produce solid PbSO_4 which is deposited at cathode plates.

The net chemical change at anode and cathode during discharge may be written as.

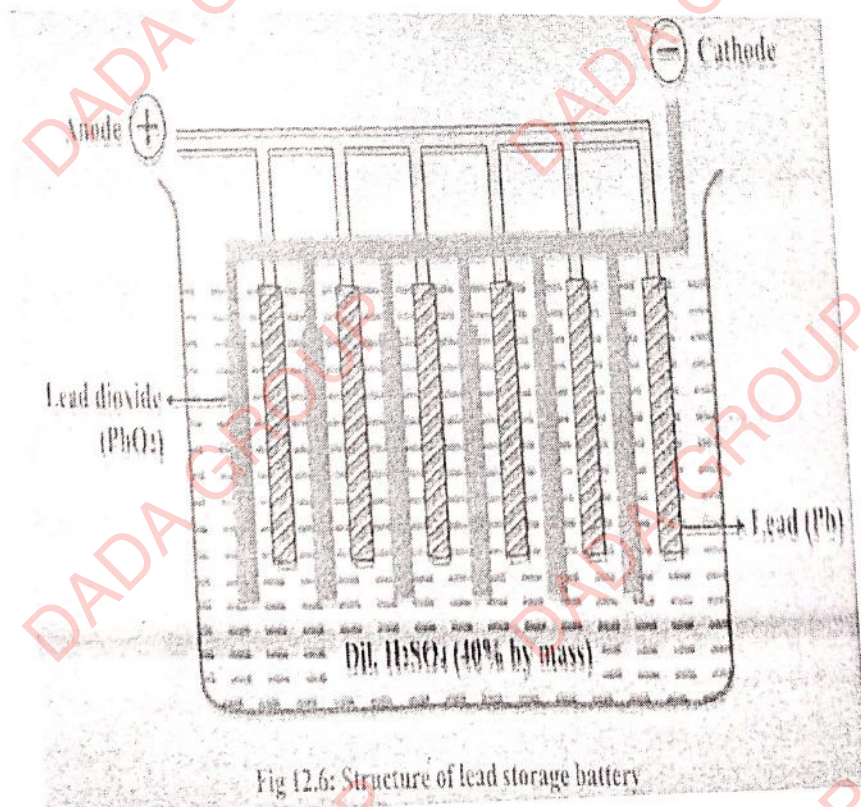
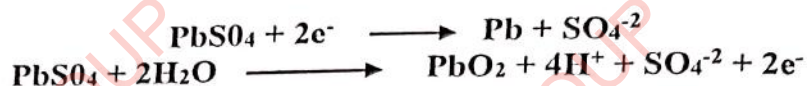


Fig 12.6: Structure of lead storage battery

As long as lead battery works, its electrolyte (H_2SO_4) uses up gradually which results in covering the electrodes with PbSO_4 . Thus it becomes unable to release electricity and said to be discharge. This can be checked by lowering in the specific gravity of acid as well as drop of voltage.

How is car battery recharged?

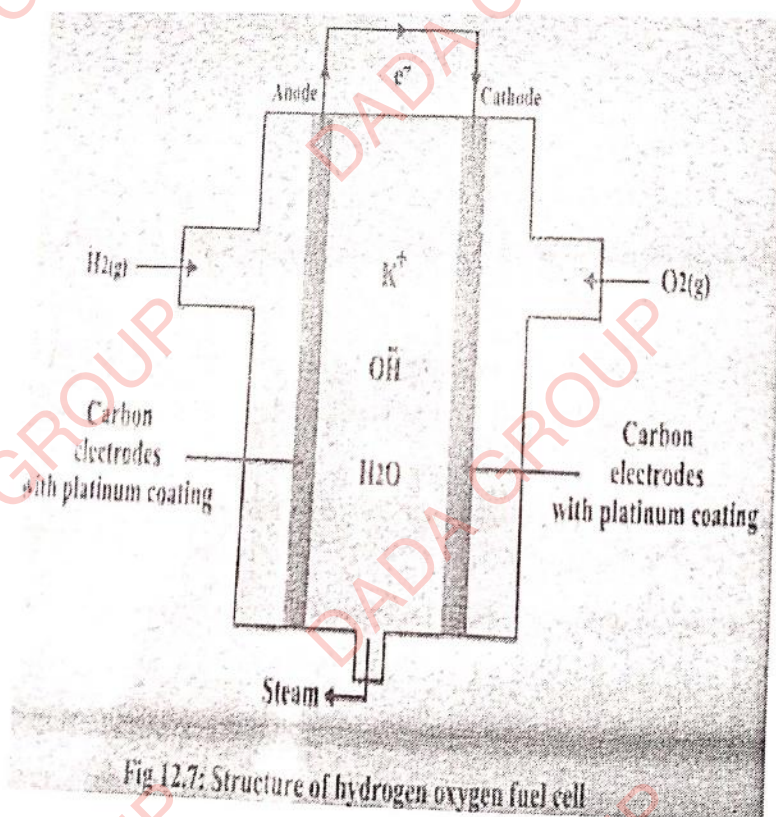
When the battery is needed to be recharged, its anode (Pb) is connected to negative terminal and cathode (PbO_2) is connected to positive terminal of external power supply. In doing so, the redox reaction becomes reverse and may be written as,



Thus H^+ ions conc. increases gradually in the electrolyte until the specific gravity again reaches the original value and the voltage of the battery is restored.

FUEL CELL:

A hydrogen oxygen fuel cell is unique type of Galvanic cell which is based on the reaction between hydrogen and oxygen to produce water and the heat release in the reaction is use to produce electricity. Unlike other batteries, it cannot store reactants hence reactants are fed continuously which utilize in the reaction and the product thus formed remove out continuously. The most familiar Fuel cell is hydrogen oxygen fuel cell. It consists of two porous carbon electrodes having a thin platinum paste. Platinum serves as catalyst. The compartment between the two electrodes is filled with concentration solution of KOH which works as electrolyte. Hydrogen gas fed into the cell from anode side and oxygen from cathode side. These gases after bubbling through their concerned electrode enter into electrolytic solution.

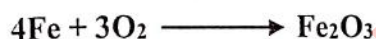


"CORROSION AND ITS PREVENTION"

A spontaneous process in which surface atoms of metals get oxidized (harmful oxides) due to the action of surrounding medium is known as corrosion."

Generally, corrosion occurs on the surface of metal but when the upper surface gets corroded, the inner layer gets exposed to air and becomes corroded. This process continues up to a certain depth which makes the metal brittle and weak. However, some metals are self-protective to corrosion since only the atoms of the outer layer oxidize.

The most familiar type of corrosion which we see in daily life is the "*rusting of iron*". Rust is a brown powdery material produced on the surface of iron appliances due to an oxidation reaction of iron atoms with moist air.



PREVENTION AGAINST CORROSION:

To keep them for a long period, they should be protected against corrosion by using certain temporary or permanent techniques.

- Paint
- Coating of grease or oil
- Coating of Metals such as zinc, tin or chromium
- Alloying of metals

GALVANIZING:

"The process of coating a thin layer of zinc on a metal is known as galvanizing."

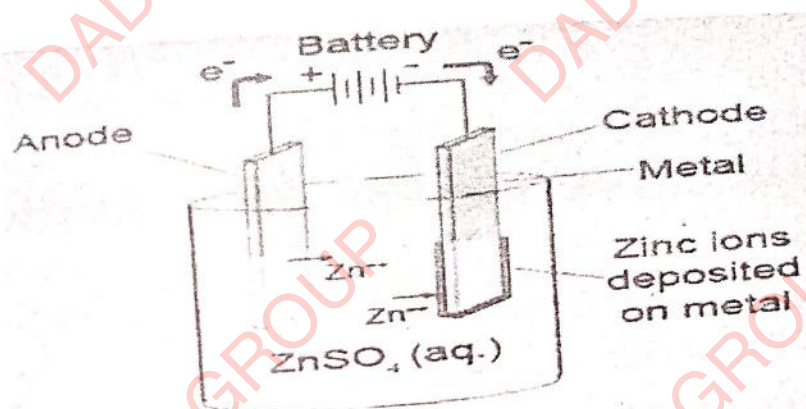


Fig 12.9: Galvanizing of metal sheet

Since zinc is placed above iron in ECS, it preferably oxidizes and consumes rather than iron and thus protects the iron against corrosion. Galvanizing is carried out either by spraying molten zinc on the surface of iron-made material or by dipping the iron sheets into molten zinc.

ELECTROPLATING:

"Electroplating is the coating of one metal on the surface of another metal by involving electrolysis"

In the electrolytic process, the iron sheets after cleaning, washing and drying are suspended into a bath containing a mixture of stannous chloride and hydrochloric acid. Pure tin is made the anode and the iron sheet works as cathode. On passing the current through the bath, tin starts depositing on the iron sheet and thus iron sheets are tinned electrolytically.

MULTIPLE CHOICE QUESTIONS

- i. The loss of electron during a chemical reaction is known as _____.
(Oxidation, Reduction, Neutralization)
- ii. The gain of electron during a chemical reaction is known as _____.
(Oxidation, Reduction, Neutralization)
- iii. The ions, which are attracted towards the anode, are known as _____.
(Anions, Cations, Positron.)
- iv. A current of one ampere flowing for one minute is equal to _____.
(One coulomb, 60 coulomb, one Faraday)
- v. A substance, which does not allow electricity to pass through, is known as _____.
(Insulator, Conductor, Electrolyte)
- vi. Such substances, which allow electricity to pass through them and are chemically decomposed, are called _____.
(Electrolytes, Insulators, Metallic conductors)
- vii. Sulphur has the highest oxidation number in _____.
(SO_2 , H_2SO_4 , H_2SO_3)
- viii. The increase of oxidation number is known as _____.
(Oxidation, Reduction, Hydrolysis)
- ix. The decrease of Oxidation number is known as _____.
(Oxidation, Reduction, Electrolysis)

THEORY QUESTIONS

SHORT QUESTION:

1. Define the following:

- Primary and secondary cell
- Oxidizing agent and reducing agent
- Oxidation number
- Oxidation and reduction
- Redox reaction

2. What is meant by oxidation number? Determine the oxidation number in following.

- Cr in H_2CrO_4
- Fe in Fe_3O_4
- S in $\text{K}_2\text{S}_4\text{O}_6$
- Cl in HClO_4

3. Define the term standard electrode potential and hydrogen electrode.

4. In the process of electroplating the item to be plated is made cathode, give reason?

5. What is corrosion? What causes it to form? What can be done to prevent its formation?

6. How can you define an electrochemical series? Give its properties.

LONG QUESTION:

- a) What is meant by electrode potential and standard Hydrogen electrode (SHE).
- b) Describe how the electrode potential of copper is determined.

2. Sketch a copper-zinc Galvanic cell, write the cell reaction and show the direction of electron flow.

3. What is the difference between a primary cell and secondary cell sketch a diagram of dry cell and explain its working.

NUMERICALS FROM PAST PAPER

OXIDATION NUMBER:

- Find oxidation number of :
a. Mn in MnO_4^- b. N in NCl_3 c. Fe in $\text{Fe}_2(\text{SO}_4)_3$ d. P in H_3PO_4
- Find the oxidation number of :
a. Cr in $\text{K}_2\text{C}_2\text{O}_7$ b. S in $\text{Na}_2\text{S}_2\text{O}_3$
- Find the oxidation number of oxygen in OF_2
- Find the oxidation number of H in H_2 .
- Find the oxidation number of the following:
a. Cl in $\text{Mg}(\text{ClO}_4)_2$ b. P in $\text{P}_2\text{O}_7^{4-}$
- Give the oxidation number of Cr in $\text{Cr}_2\text{O}_7^{2-}$

EQUATION BALANCING (ION ELECTRON METHOD):

- $\text{Cl}_2 + \text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}_3^- + \text{H}_2\text{O}$ (Basic)
- $\text{Cr}_2\text{O}_7^{2-} + \text{I}_2 \rightarrow \text{Cr}^{3+} + \text{IO}_3^-$ (Acidic)
- Balance the following equation by ion electron method:
 - $\text{Cr}(\text{OH})_3 + \text{SO}_4^{2-} \rightarrow \text{CrO}_4^{2-} + \text{SO}_3^{2-}$ (Basic medium)

$$2\text{Cr}(\text{OH})_3 + 3\text{SO}_4^{2-} + 4\text{OH}^- \rightarrow 2\text{CrO}_4^{2-} + 3\text{SO}_3^{2-} + 5\text{H}_2\text{O}$$
 - $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} \rightarrow \text{Mn}^{2+} + \text{CO}_2$ (acidic medium)

$$2\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O}$$
- $\text{MnO}_4^- + \text{SO}_3^{2-} \rightarrow \text{Mn}^{2+} + \text{SO}_4^{2-}$ (Basic)

$$2\text{MnO}_4^- + 5\text{SO}_3^{2-} + 3\text{H}_2\text{O} \rightarrow 2\text{Mn}^{2+} + 5\text{SO}_4^{2-} + 6\text{OH}^-$$
- $\text{MnO}_4^- + \text{Cl}^- \rightarrow \text{Mn}^{2+} + \text{Cl}_2$ (acidic)
- $\text{MnO}_4^- + \text{SO}_3^{2-} \rightarrow \text{MnO}_4^{2-} + \text{SO}_4^{2-}$ (basic)

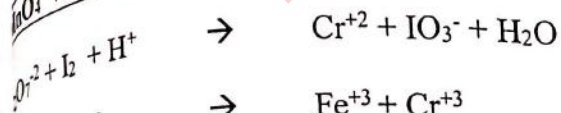
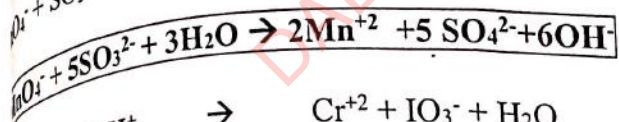
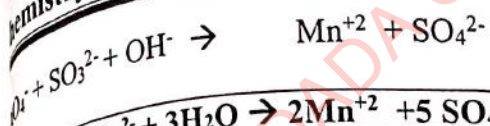
$$2\text{MnO}_4^- + \text{SO}_3^{2-} + 2\text{OH}^- \rightarrow 2\text{MnO}_4^{2-} + \text{SO}_4^{2-} + \text{H}_2\text{O}$$
- $\text{MnO}_4^- + \text{Fe}^{2+} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+} + \text{H}_2\text{O}$
- $\text{MnO}_4^- + \text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+}$ (acidic)
- $\text{Cr}_2\text{O}_7^{2-} + \text{I}^- \rightarrow \text{Cr}^{3+} + \text{I}_2$ (acidic)

$$\text{Cr}_2\text{O}_7^{2-} + 6\text{I}^- + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 3\text{I}_2 + 7\text{H}_2\text{O}$$
- $\text{Cr}_2\text{O}_7^{2-} + \text{I}_2 \rightarrow \text{Cr}^{3+} + \text{IO}_3^-$ (acidic)

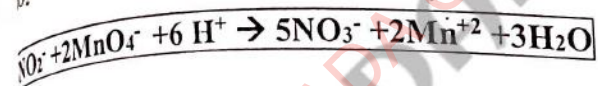
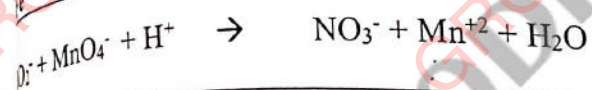
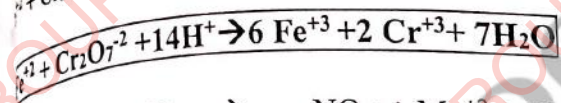
ELECTROCHEMISTRY

chapter 12

Chemistry I



(acidic)



(basic)



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